

Chapter 4: Matrices and matrix operations Worksheet

1. Are each of the following statements true or false?

- (a) The transpose of a 2×4 matrix is a 4×2 matrix.
- (b) If A, B, C are 3×3 matrices with $A + B = A + C$, then $B = C$.
- (c) If $A\mathbf{x} = \mathbf{0}$ where $\mathbf{x} \neq \mathbf{0}$, then A must be the zero matrix.
- (d) If $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ in terms of its columns, and if $\mathbf{b} = 3\mathbf{a}_1 - 2\mathbf{a}_2$, then the system $A\mathbf{x} = \mathbf{b}$ has a solution.
- (e) If A is $m \times n$ matrix and $m < n$, then $A\mathbf{x} = \mathbf{b}$ has a solution for every column vector $\mathbf{b}$.
- (f) If the products AB and BA are both possible, then A, B are both square matrices (i.e. $n \times n$ for some n).
- (g) If A is a 7×3 matrix, then AA^T is a 7×7 matrix.
- (h) If A, B, C are 3×3 matrices with $AB = AC$ and $A \neq O$, then $B = C$.

Solution.

- (a) True. Generally, the transpose of an $m \times n$ matrix is an $n \times m$ matrix.
- (b) True. Add $-A$ to both sides.
- (c) False. ex: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$
- (d) True. Take $\mathbf{x} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$
- (e) False. Take the matrix A to the zero matrix, and $\mathbf{b}$ has non-zero elements. How to see this: look at all possible reduced row echelon forms.
- (f) False. A could be 2×3 , and B be 3×2 . Then AB and BA are both possible (AB is 2×2 and BA is 3×3).
- (g) True, A^T will be 3×7 , so the product AA^T is 7×7 .

- (h) False, you can't necessarily cancel A . For example, if $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$,
$$B = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 3 & 0 & 2 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ -4 & 4 & 4 \end{bmatrix}, AB = AC = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ but } B \neq C.$$

2. Let $A = \begin{bmatrix} 2 & -4 & 5 \\ 3 & 1 & -2 \end{bmatrix}$. What is a_{11}, a_{22}, a_{23} ?

Solution.

$a_{11} = 2, a_{22} = 1, a_{23} = -2$.

3. Define the following matrices $A = \begin{bmatrix} 2 & -4 & 0 \\ 3 & 0 & -2 \\ 0 & 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}$, $C = \begin{bmatrix} 4 & -2 \\ 3 & -5 \end{bmatrix}$,

$$D = \begin{bmatrix} 0 & 5 & 4 \\ 1 & 7 & -2 \\ -1 & 2 & 3 \end{bmatrix}, \quad E = \begin{bmatrix} -5 & -2 & 1 \\ 3 & -3 & 2 \end{bmatrix}, \quad F = \begin{bmatrix} 3 & 2 \\ 4 & -5 \\ 0 & 1 \end{bmatrix}$$

Perform each calculation, if possible. If not possible, explain.

- (a) $A + B$
- (b) $A + 2D$
- (c) $3C + 2B + E$
- (d) $C - C^T$
- (e) $E^T + F$

Solution.

- (a) This expression is not valid because A is a 3×3 matrix, whereas B is a 3×1 matrix.

$$(b) \quad A + 2D = \begin{bmatrix} 2 & 6 & 8 \\ 5 & 14 & -6 \\ -2 & 4 & 3 \end{bmatrix}$$

- (c) This expression is not valid because B is a 3×1 matrix, C is a 2×2 matrix, and E is a 2×3 matrix.

$$(d) \quad C - C^T = \begin{bmatrix} 4 & -2 \\ 3 & -5 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ -2 & -5 \end{bmatrix} = \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix}$$

$$(e) \quad E^T + F = \begin{bmatrix} -5 & 3 \\ -2 & -3 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ 4 & -5 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 5 \\ 2 & -8 \\ 1 & 3 \end{bmatrix}$$

4. Find the matrix A given the equation below.

$$\left(4A - 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right)^T = \begin{bmatrix} 0 & 1 \\ 2 & 4 \end{bmatrix}$$

Solution.

Remember that for matrices A, B and scalar k , $(A + B)^T = A^T + B^T$, and $(kA)^T = kA^T$, and $(A^T)^T = A$. We can use these facts to solve for A . (NOTE: There are multiple ways to approach this problem). I take the transpose of both sides:

$$\left(\left(4A - 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right)^T\right)^T = \begin{bmatrix} 0 & 1 \\ 2 & 4 \end{bmatrix}^T$$

$$4A - 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

$$4A = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 8 \end{bmatrix}$$

$$A = \frac{1}{4} \begin{bmatrix} 2 & 2 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & 2 \end{bmatrix}$$

5. Given A and $\mathbf{x}$ below, give the size/dimension of A , $\mathbf{x}$, and $A\mathbf{x}$. Compute $A\mathbf{x}$ using both notions of matrix-vector multiplication: (a) as a linear combination of the columns of A and (b) as repeated dot-product.

$$A = \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 0 & 1 & 4 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 4 \end{bmatrix}$$

Solution.

A is 2×4 , $\mathbf{x}$ is 4×1 , so $A\mathbf{x}$ will be 2×1 . Using linear combination:

$$A\mathbf{x} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} 29 \\ 16 \end{bmatrix}$$

Using the dot-product:

$$A\mathbf{x} = \begin{bmatrix} (1)(1) + (2)(2) + (0)(0) + (6)(4) \\ (0)(1) + (0)(2) + (1)(0) + (4)(4) \end{bmatrix} = \begin{bmatrix} 29 \\ 16 \end{bmatrix}$$

6. Given

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} -5 & -2 & 1 \\ 3 & -3 & 2 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad F = \begin{bmatrix} 3 & 2 \\ 4 & -5 \\ 0 & 1 \end{bmatrix}$$

If possible, compute the following.

- (a) AB
- (b) AC
- (c) C^2 (or CC)
- (d) $B^T EF$
- (e) $E(D + F^T)$

Solution.

- (a) This expression is not valid because A is a 2×2 matrix, whereas B is a 3×1 matrix.
- (b) $AC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
- (c) $C^2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$
- (d) This expression is not valid because it is not possible to multiply B^T and E . B^T is a 1×3 matrix, whereas E is a 2×2 matrix.
- (e) We first compute $D + F^T$.

$$D + F^T = \begin{bmatrix} -5 & -2 & 1 \\ 3 & -3 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 4 & 0 \\ 2 & -5 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 1 \\ 5 & -8 & 3 \end{bmatrix}.$$

We then compute $E(D + F^T)$.

$$E(D + F^T) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -2 & 2 & 1 \\ 5 & -8 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 1 \\ -5 & 8 & -3 \end{bmatrix}.$$